

A step-in-ready framework that turns every requirement into a typed, machine-readable, enforceable engineering artifact — and lets your AI operate inside the control envelope your standards already define.

01 · THE GAP

Why every WG keeps hitting the same wall

Reqs vanish after build Live in PDFs and ALM tools, then forgotten — no live baseline to trace failures back to.	Terminology tribalism Same concept, three vocabularies. STPA classifies it as a formal hazard, not a nuisance.
AI lacks awareness Models see raw prompts; they cannot tell which standard they satisfied or which UCA they avoided.	Governance ≠ enforcement A standard you can't auto-evaluate is advice, not control. Silent non-compliance is the default.

02 · THE MECHANISM

The gateway — five steps, plain language

01 IDENTIFY Who is asking and what job they need done. Persona · JTBD · Domain	02 RESOLVE Eliminate ambiguity before it enters the system. Canonical ontology	03 CONSTRAIN Test structural completeness, citations, safety. SHACL shapes	04 GOVERN Inject rules — outcomes, conflicts, timing. SWRL + UCAs	05 AUTHORIZE Compile directive; humans approve irreversibles. Simplex Action Gate
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SERVICE FACADE · 10-CRITERION VALIDATION Every AI response evaluated against acceptance criteria before delivery.	SIMPLEX ACTION GATEWAY No irreversible action without a verified human handshake
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03 · THE VALUE & WHERE YOU PLUG IN

Three benefits — and a defined point of entry for every expertise

Controllability over reliability STAMP-aligned. Scales as AI capabilities grow.	4 / 4 UCA coverage Missing · Hazardous · Mis-timed · Wrong-duration.	End-to-end traceability Need → ontology → SHACL → SWRL → CAD → response.		
Domain practitioner AERO · RAIL · AUTO Bring requirements & failure modes.	Systems engineer MBSE Map JTBD → system decomp; UCAs → constraints.	Safety analyst STPA · HAZOP Define UCAs; classify Zone 6 reversibility.	Ontologist KNOWLEDGE ENG. Author vocabulary; design SHACL & SWRL.	Compliance steward POLICY · STDS Activate, validate & retire requirements.

ASK → Bring 5 contested terms · 1 real requirement · 1 interoperability failure
1 SHACL shape from your domain — for the next session.